

Machine Learning Assignment 1: Polynomial Regression Report

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<https://github.com/rajg2906/Polynomial-Regression>

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1 Introduction

This report details the methodology and implementation of polynomial regression models developed for two distinct stages of a geothermal power plant expansion project. As outlined in assignment, the objective is to accurately predict a continuous target variable, y , using polynomial regression.

2 Problem Statements

2.1 Phase 1: Power Plant Steam Turbine Optimization (var1)

In the first phase of the assignment, the role is that of a Lead Renewable Energy Engineer aiming to optimize surface energy generation prior to drilling new wells. The plant operates a multi-stage steam turbine system with unique behavioral characteristics due to variations in manufacturing tolerances and local water chemistry.

The turbine efficiency is determined by six specific operational parameters, defined as percentage deviations:

- x_1 : High-pressure steam valve adjustment
- x_2 : Condenser coolant flow rate adjustment
- x_3 : Re-injection pump hydraulic pressure
- x_4 : Turbine blade pitch angle
- x_5 : Non-condensable gas exhaust valve rate
- x_6 : Steam inlet pressure adjustment

The target variable is the Net Power Score (y), which can be modeled using a polynomial up to degree 10 based on these parameters. Historical plant calibration logs are used to train the model.

2.2 Phase 2: Subterranean Thermal Reservoir Mapping (var2)

The second phase focuses on the selection of precise coordinates for drilling new geothermal extraction wells within a high-potential geological survey block. Seismic and thermal probe sensors deployed across a 3D grid record data based on spatial offsets from a central basecamp:

- x_1 : East-West coordinate offset in meters
- x_2 : North-South coordinate offset in meters
- x_3 : Vertical depth offset relative to the basecamp in meters

The target variable here is the Thermal Anomaly Score (y) recorded at each coordinate. Complex geological structures dictate that the heat map resembles a high-degree polynomial, potentially up to degree 20. The objective is to predict the Thermal Anomaly Score across planned drilling sites using a model trained on a random scatter of physical core samples.

3 Methodology

3.1 Data Preprocessing and Pipeline Architecture

To handle the scale and complexity of high-degree polynomial regression, a structured machine learning pipeline was implemented using Python's scikit-learn library. The core steps of the pipeline are as follows:

1. **Standardization:** Polynomial features grow exponentially in magnitude, which can easily lead to numerical overflow and model instability. `StandardScaler` was applied immediately to center the features to a mean of zero and scale them to a variance of one.
2. **Polynomial Feature Generation:** The `PolynomialFeatures` class was utilized to generate interacting terms up to a specified degree d . As per the assignment guidelines, a polynomial of degree x ensures that for each term, the sum of the powers of features adds at most up to x .
3. **Regularized Regression:** Unconstrained high-degree polynomial regression is heavily prone to overfitting. To mitigate this, L_1 (Lasso) and L_2 (Ridge) regularized regression models were implemented.

3.2 Cross-Validation Strategy

Model evaluation heavily relied on a K-Fold cross-validation strategy to ensure robust generalization to unseen data. A 5-fold cross-validation splitter (`KFold` with `n_splits=5`, shuffled with `random state=101`) was employed. This effectively partitioned the training

data into an 80-20 split iteratively, training on 80% and validating on 20% across five folds.

4 Model Selection and Evaluation Metrics

4.1 Evaluation Metrics

The models were evaluated against hidden ground truth data using two primary metrics:

- **Mean Squared Error (MSE):** Measures the average squared difference between the predicted values and the actual values. It penalizes larger errors severely.
- **R^2 Score (Coefficient of Determination):** Evaluates how well the polynomial model captures the variance of the target variable.

4.2 Phase 1 (var1): Lasso (L_1) Regression

For the Steam Turbine Optimization task, which features 6 input variables, generating polynomials up to degree 10 creates a massive feature space. Consequently, Lasso Regression (LassoCV) was chosen. Lasso applies an L_1 penalty, which naturally drives the coefficients of less important polynomial terms exactly to zero, performing implicit feature selection. The hyperparameter α was tuned using internal 3-fold cross-validation with a maximum of 2500 iterations to ensure convergence.

4.3 Phase 2 (var2): Ridge (L_2) Regression

For the Subterranean Thermal Reservoir Mapping task, which features 3 input variables but requires polynomials up to degree 20, Ridge Regression (RidgeCV) was utilized. Ridge applies an L_2 penalty, continuously shrinking coefficient magnitudes without eliminating them. Because thermal mapping relies on continuous spatial interactions where extreme localized spikes must be smoothed out, Ridge's tendency to distribute the penalty across all correlated spatial features proved more effective. An explicit grid of penalty values ($\alpha \in [0.01, 0.1, 1.0, 10.0, 100.0]$) was evaluated.

5 Rationale for Polynomial Degree Selection

The optimal polynomial degree chosen for Phase-1 was 5 and for Phase-2 it was 10.

The optimal polynomial degree was dynamically selected through the custom execute hybrid pipeline function. The logic involved iterating through all possible degrees. For each degree, the full modeling pipeline was cross-validated.

The rationale for the final chosen degree was explicitly based on maximizing the crossvalidated R^2 score while monitoring the MSE to ensure the model avoided underfitting or overfitting uniquely generated student data. The degree that yielded the highest mean test R^2 across the 5 folds was locked in as best deg.

Once the optimal degree was identified, the final pipeline was reconstructed using this specific degree and retrained on the entirety of the training set to maximize predictive power before inferring the test set targets.

6 Results

```
*** >>> Initiating Lasso (L1) Regression for variant 1 <<<
Eval Degree 1: R2 = 0.0828 | MSE = 9.0832
Eval Degree 2: R2 = 0.6938 | MSE = 3.0090
Eval Degree 3: R2 = 0.8985 | MSE = 1.0000
Eval Degree 4: R2 = 0.9359 | MSE = 0.6224
Eval Degree 5: R2 = 0.9633 | MSE = 0.3578
Eval Degree 6: R2 = 0.9586 | MSE = 0.4010
Eval Degree 7: R2 = 0.9508 | MSE = 0.4756
Eval Degree 8: R2 = 0.9443 | MSE = 0.5344
Eval Degree 9: R2 = 0.9264 | MSE = 0.7041
Eval Degree 10: R2 = 0.9154 | MSE = 0.8140

=> Best Lasso (L1) Model for var1: Degree 5 (R2: 0.9633, MSE: 0.3578)

=> Successfully generated BT2024172-pred_var1.csv

>>> Initiating Ridge (L2) Regression for variant 2 <<<
Eval Degree 1: R2 = 0.2587 | MSE = 34.0358
Eval Degree 2: R2 = 0.5210 | MSE = 21.7896
Eval Degree 3: R2 = 0.7592 | MSE = 10.7876
Eval Degree 4: R2 = 0.9097 | MSE = 4.0224
Eval Degree 5: R2 = 0.9663 | MSE = 1.5284
Eval Degree 6: R2 = 0.9857 | MSE = 0.6296
Eval Degree 7: R2 = 0.9921 | MSE = 0.3620
Eval Degree 8: R2 = 0.9938 | MSE = 0.2816
Eval Degree 9: R2 = 0.9937 | MSE = 0.2859
Eval Degree 10: R2 = 0.9941 | MSE = 0.2674
Eval Degree 11: R2 = 0.9938 | MSE = 0.2811
Eval Degree 12: R2 = 0.9935 | MSE = 0.2906
Eval Degree 13: R2 = 0.9927 | MSE = 0.3265
Eval Degree 14: R2 = 0.9924 | MSE = 0.3413
Eval Degree 15: R2 = 0.9913 | MSE = 0.3912
Eval Degree 16: R2 = 0.9902 | MSE = 0.4379
Eval Degree 17: R2 = 0.9876 | MSE = 0.5583
Eval Degree 18: R2 = 0.9846 | MSE = 0.6944
Eval Degree 19: R2 = 0.9755 | MSE = 1.1240
Eval Degree 20: R2 = 0.9648 | MSE = 1.6407

=> Best Ridge (L2) Model for var2: Degree 10 (R2: 0.9941, MSE: 0.2674)

=> Successfully generated BT2024172-pred_var2.csv
```

7 Conclusion

By leveraging regularized polynomial regression within a standardized, cross-validated pipeline, the models successfully capture the complex non-linear relationships present in the geothermal datasets. The use of Lasso regression effectively navigated the dense feature space of the 6-variable surface plant parameters, while Ridge regression provided the necessary smoothing for the high-degree 3D subterranean spatial mapping. The finalized predictions have been saved to the required .csv formats.